Meeting 4D — Runtime State Architecture

Part 4: Loader, Background Services, Button & Error State Machines

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Purpose

This document defines the remaining **system-level runtime state machines** that keep the workstation operating correctly behind the scenes.

Unlike the Voice Engine (which will be discussed separately), these states are **true firmware runtime states**. They exist continuously while the ESP32 is running and coordinate loading, scanning, monitoring, and fault recovery.

Runtime State Machine 9 — Instrument Loader

The Instrument Loader manages loading VHL instruments from SD card into RAM.

Only one instrument may be actively loaded at a time.

IL1 — LOADER_IDLE

Purpose

Instrument loader inactive.

Entry

System reaches normal operation.

Responsibilities

- Wait for instrument selection.
- Keep currently loaded instrument active.

Valid Transition

→ LOADER_PREPARE

Exit

User selects another instrument.

IL2 — LOADER_PREPARE

Purpose

Prepare instrument loading.

Entry

New instrument selected.

Responsibilities

- Stop current voices.
- Pause accompaniment if required.
- Reserve RAM.

Valid Transition

→ LOADER_READING

Exit

RAM prepared.

IL3 — LOADER_READING

Purpose

Read VHL file from SD card.

Entry

Preparation complete.

Responsibilities

- Open VHL file.
- Read binary contents.
- Verify file integrity.

Valid Transition

→ LOADER_VERIFY

Exit

Entire file loaded.

IL4 — LOADER_VERIFY

Purpose

Validate loaded instrument.

Entry

File completely read.

Responsibilities

- Verify CRC
- Verify header
- Verify memory layout
- Verify sample references

Valid Transition

→ LOADER_INSTALL

→ LOADER_ERROR

Exit

Validation completed.

IL5 — LOADER_INSTALL

Purpose

Activate the newly loaded instrument.

Entry

Verification successful.

Responsibilities

- Replace previous instrument.
- Build runtime lookup tables.
- Reset voice allocator.

Valid Transition

→ LOADER_COMPLETE

Exit

Installation finished.

IL6 — LOADER_COMPLETE

Purpose

Return workstation to normal operation.

Entry

Installation successful.

Responsibilities

- Update UI.
- Display instrument name.
- Enable keyboard.

Valid Transition

→ LOADER_IDLE

Exit

User resumes playing.

IL7 — LOADER_ERROR

Purpose

Recover from failed instrument loading.

Entry

Verification failed.

Responsibilities

- Discard partial data.
- Restore previous instrument.
- Notify user.

Valid Transition

→ LOADER_IDLE

Exit

Recovery complete.

Runtime State Machine 10 — Button Manager

The Button Manager controls all dedicated control buttons independently of keyboard scanning.

B1 — BUTTON_IDLE

Purpose

Wait for button activity.

Entry

System running.

Responsibilities

- Scan buttons.
- Detect state changes.

Valid Transition

→ BUTTON_DEBOUNCE

Exit

Potential press detected.

B2 — BUTTON_DEBOUNCE

Purpose

Remove switch bounce.

Entry

Button edge detected.

Responsibilities

- Wait debounce interval.
- Verify stable signal.

Valid Transition

→ BUTTON_PRESSED

→ BUTTON_IDLE

Exit

Button confirmed.

B3 — BUTTON_PRESSED

Purpose

Generate button press event.

Entry

Debounce successful.

Responsibilities

- Notify Runtime Manager.
- Start hold timer.

Valid Transition

→ BUTTON_HELD

→ BUTTON_RELEASED

Exit

Button changes state.

B4 — BUTTON_HELD

Purpose

Handle long button presses.

Entry

Hold threshold exceeded.

Responsibilities

- Generate repeat events if required.
- Continue monitoring.

Valid Transition

→ BUTTON_RELEASED

Exit

Button released.

B5 — BUTTON_RELEASED

Purpose

Finish button interaction.

Entry

Button released.

Responsibilities

- Stop timers.
- Clear temporary state.

Valid Transition

→ BUTTON_IDLE

Exit

Ready for next press.

Runtime State Machine 11 — Background Service Manager

These services continuously execute while the workstation is operating.

BG1 — BACKGROUND_IDLE

Purpose

Background scheduler waiting.

Entry

Normal operation.

Responsibilities

- Schedule periodic services.

Valid Transition

→ BACKGROUND_RUNNING

Exit

Scheduler tick.

BG2 — BACKGROUND_RUNNING

Purpose

Execute low-priority maintenance tasks.

Entry

Scheduler activated.

Responsibilities

- Refresh LCD
- Update statistics
- Poll SD status
- Refresh battery status (future)

Valid Transition

→ BACKGROUND_IDLE

Exit

Tasks complete.

Runtime State Machine 12 — Storage Manager

Controls SD card access.

ST1 — STORAGE_IDLE

Purpose

No storage operation.

Entry

System ready.

Responsibilities

Wait for requests.

Valid Transition

→ STORAGE_READING

→ STORAGE_WRITING

Exit

Storage requested.

ST2 — STORAGE_READING

Purpose

Read from SD card.

Entry

Read requested.

Responsibilities

- Read requested blocks.
- Verify transfer.

Valid Transition

→ STORAGE_IDLE

→ STORAGE_ERROR

Exit

Transfer complete.

ST3 — STORAGE_WRITING

Purpose

Write data to SD card.

Entry

Write requested.

Responsibilities

- Store recording.
- Flush buffers.
- Verify completion.

Valid Transition

→ STORAGE_IDLE

→ STORAGE_ERROR

Exit

Write completed.

ST4 — STORAGE_ERROR

Purpose

Recover from SD failures.

Entry

Read/write failed.

Responsibilities

- Retry operation.
- Notify Runtime Manager.
- Preserve system stability.

Valid Transition

→ STORAGE_IDLE

Exit

Recovery finished.

Runtime State Machine 13 — Error Manager

Centralized runtime fault handling.

E1 — ERROR_IDLE

Purpose

Normal operation.

Entry

No active faults.

Responsibilities

Monitor system.

Valid Transition

→ ERROR_DETECTED

Exit

Fault detected.

E2 — ERROR_DETECTED

Purpose

Capture runtime fault.

Entry

Subsystem reports error.

Responsibilities

- Log fault.
- Identify severity.
- Select recovery action.

Valid Transition

→ ERROR_RECOVERY

→ ERROR_FATAL

Exit

Classification complete.

E3 — ERROR_RECOVERY

Purpose

Attempt automatic recovery.

Entry

Recoverable fault.

Responsibilities

- Reset subsystem.
- Retry operation.
- Restore normal execution.

Valid Transition

→ ERROR_IDLE

→ ERROR_FATAL

Exit

Recovery successful or failed.

E4 — ERROR_FATAL

Purpose

Protect system integrity.

Entry

Unrecoverable error.

Responsibilities

- Stop audio.
- Disable keyboard.

- Display error message.

Valid Transition

→ SHUTDOWN

Exit

System restart required.

Summary

This document defines **28 additional runtime states**.

Instrument Loader

- LOADER_IDLE
- LOADER_PREPARE
- LOADER_READING
- LOADER_VERIFY
- LOADER_INSTALL
- LOADER_COMPLETE
- LOADER_ERROR

Button Manager

- BUTTON_IDLE
- BUTTON_DEBOUNCE
- BUTTON_PRESSED
- BUTTON_HELD
- BUTTON_RELEASED

Background Service Manager

- BACKGROUND_IDLE
- BACKGROUND_RUNNING

Storage Manager

- STORAGE_IDLE
- STORAGE_READING
- STORAGE_WRITING
- STORAGE_ERROR

Error Manager

- ERROR_IDLE
- ERROR_DETECTED
- ERROR_RECOVERY
- ERROR_FATAL

Together with Meetings 4A–4C, these complete the firmware's high-level runtime state architecture. The remaining discussion will focus on the **internal voice processing architecture** (voice allocation, sample playback, envelopes, filters, modulation, mixer, DSP, and audio pipeline), which are processing pipelines rather than global firmware states.